ECE 657D: Research Project

The Impact of Prior Geometry on VAE Latent Space and Generation

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Abstract

The latent prior in a variational autoencoder (VAE) shapes the latent geometry and generation quality, yet most models default to an isotropic Gaussian prior. We present a strictly controlled comparison of five priors—Gaussian, Laplace, Student-t, mixture of Gaussians (MoG), and VampPrior—on Fashion-MNIST. Our results show that despite differing tail behaviors, the three unimodal priors perform similarly across all metrics. In contrast, the multimodal priors reduce FID by ∼15 points, more than halve KID, and yield better-organized latent spaces with more active units and tighter class clusters. We conclude that prior multimodality, rather than merely non-Gaussianity, is the decisive factor for improving VAE generation.

Introduction

Variational autoencoders (VAEs) regularize their latent space through a prior distribution [Kingma and Welling, 2013]. The standard isotropic Gaussian choice is convenient but often poorly matched to the aggregate posterior $q_\phi(\mathbf{z}) = \frac{1}{N}\sum_i q_\phi(\mathbf{z}|\mathbf{x}_i)$ induced over the dataset. This mismatch causes prior samples to fall in low-density regions, yielding unrealistic outputs [Bauer and Mnih, 2019, Aneja et al., 2021]. Alternatives like learnable mixtures [Dilokthanakul et al., 2016], VampPrior [Tomczak and Welling, 2018], and heavy-tailed distributions [Abiri and Ohlsson, 2020] mitigate this, but are typically evaluated alongside architectural or objective changes, obscuring the isolated impact of the prior.

We present a controlled comparison of five priors—Gaussian, Laplace, Student-t, mixture of Gaussians (MoG), and VampPrior—on Fashion-MNIST [Xiao et al., 2017], keeping the architecture, objective, and hyperparameters strictly fixed. Our results reveal a clear performance divide. Multimodal priors (MoG, VampPrior) substantially outperform all three unimodal priors on every generation metric, while the unimodal priors perform indistinguishably from one another. We demonstrate this gap stems from latent-space organization: multimodal priors activate more dimensions and produce tighter per-class clusters.

Related Work

VAEs traditionally employ an isotropic Gaussian prior [Kingma and Welling, 2013]. Research addressing the prior–aggregate-posterior gap generally falls into two categories. *Fixed replacements*, like Laplace or Student-t priors [Abiri and Ohlsson, 2020], alter tail behavior without introducing new parameters. *Learnable priors*—such as Gaussian mixtures [Dilokthanakul et al., 2016], VampPrior

[Tomczak and Welling, 2018], and resampled variants [Bauer and Mnih, 2019, Aneja et al., 2021]—adapt to the data distribution to improve sample quality. A concurrent issue is posterior collapse, often diagnosed by counting active latent units [He et al., 2019]. While prior choice likely influences collapse severity, its isolated effect lacks controlled empirical study—the precise gap this project addresses.

Method

Model and objective. We train a convolutional VAE on Fashion-MNIST [Xiao et al., 2017] with a 16-dimensional latent space. Models minimize $\mathcal{L} = \mathrm{BCE}(\mathbf{x}, \hat{\mathbf{x}}) + \mathrm{KL}[q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})]$ using Adam optimizer (lr=10^{-3}), batch size 1024, and 1000 epochs. While the Gaussian KL divergence uses its standard closed form, we use a single-sample Monte Carlo estimator $\mathrm{KL} \approx \log q_\phi(\mathbf{z}|\mathbf{x}) - \log p(\mathbf{z})$ for all other priors.

Priors. Keeping all other components fixed, we compare five priors $p(\mathbf{z})$. Three are unimodal and zero-centered: (1) *Gaussian*: $\prod_j \mathcal{N}(0,1)$; (2) *Laplace*: $\prod_j \mathrm{Lap}(0,1)$; and (3) *Student-t*: $\prod_j t(\nu{=}3)$ [Abiri and Ohlsson, 2020]. Two are multimodal and learnable: (4) *MoG*: a 500-component mixture [Dilokthanakul et al., 2016]; and (5) *VampPrior*: using 500 learnable pseudo-inputs [Tomczak and Welling, 2018].

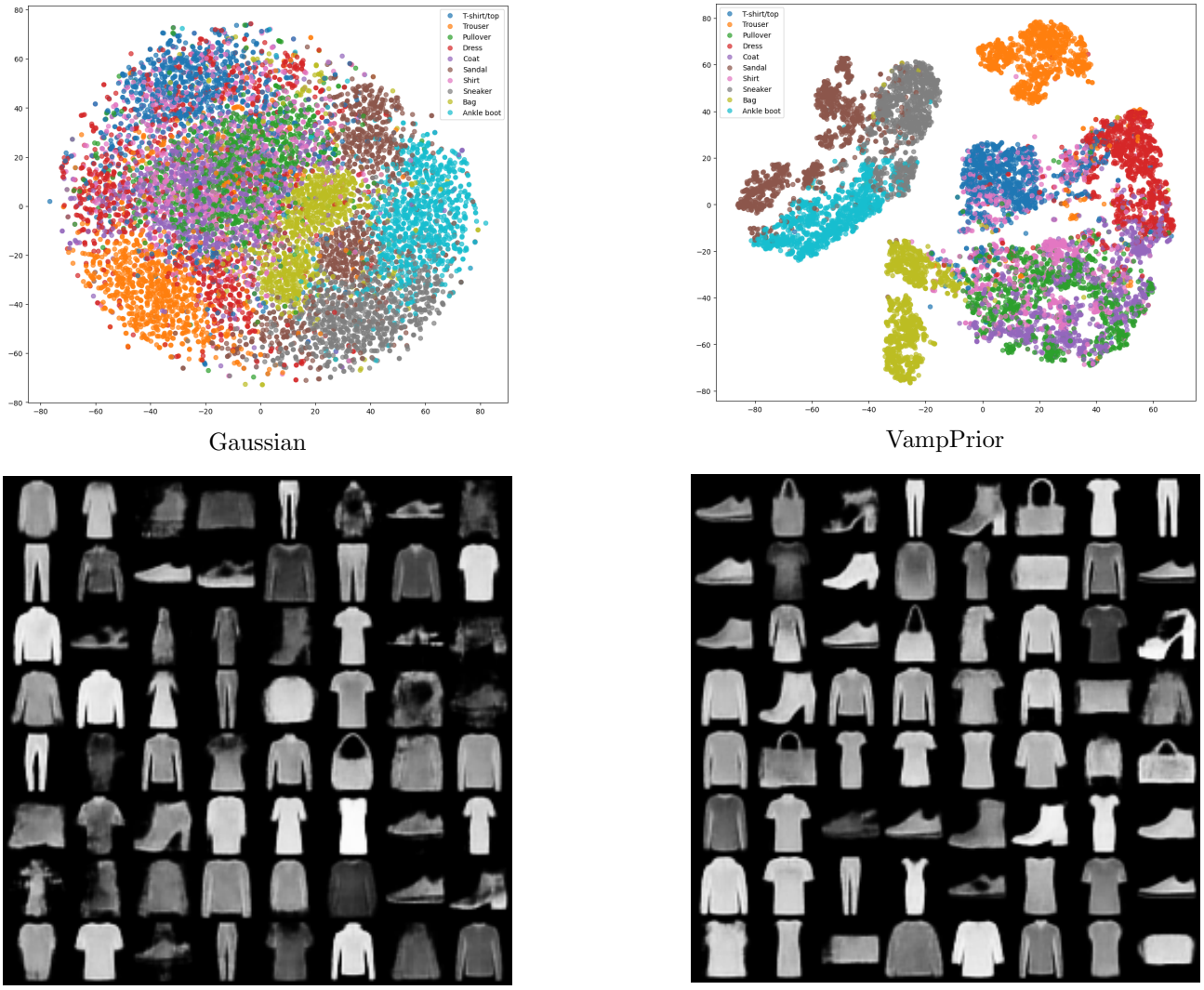
Evaluation. We measure generation quality via FID [Heusel et al., 2017] and KID [Bińkowski et al., 2018] on prior samples, density estimation via importance-sampled marginal log-likelihood (K=5000), and posterior collapse via active latent units (variance > 0.01) [He et al., 2019].

Experiments

Table 1 reveals two distinct performance tiers. Unimodal priors (Gaussian, Laplace, Student-t) cluster together: FID spans 49.5–50.9, KID varies by <0.003, and log-likelihoods fall within 0.6. Changing tail shape yields no measurable effect on generation, latent utilization, or clustering. Conversely, multimodal priors (MoG, VampPrior) form a clearly superior tier, reducing FID by ∼15 points and more than halving KID. VampPrior achieves the best FID (35.53) and KID (.0188). This confirms multimodality—not merely non-Gaussianity—is the decisive factor.

Table 1: Main results. FID and KID are computed on prior samples. $\log p(\mathbf{x})$: test marginal log-likelihood via importance sampling (K=5000). AU: active units out of 16 (variance threshold 0.01). Best in **bold**.

Prior	**FID** ↓	**KID** ↓	$\log p(\mathbf{x})$ ↑	**AU** ↑
Gaussian	50.09	.0392 ± .0019	−233.20	9 / 16
Laplace	49.45	.0386 ± .0020	−233.74	8 / 16
Student-t	50.85	.0410 ± .0021	−233.73	8 / 16
MoG	38.46	.0209 ± .0012	**−231.12**	11 / 16
VampPrior	**35.53**	**.0188** ± .0010	−231.17	**12** / 16



Gaussian VampPrior

Gaussian VampPrior

Figure 1: **Top:** t-SNE of test-set latent codes showing tighter class clustering for VampPrior. **Bottom:** Unconditional prior samples demonstrating sharper generations for VampPrior.

Latent space structure. The generation gap is mirrored in latent organization. MoG activates 11 of 16 dimensions and VampPrior activates 12 (vs. 8–9 for unimodal priors), indicating less posterior collapse. VampPrior produces the clearest class separation. Figure 1 (Top) illustrates this: the Gaussian latent space is a single diffuse cloud, while VampPrior forms visibly tighter per-class clusters (all five plots in Appendix B).

Prior–aggregate-posterior alignment. Comparing Table 1 with Table 2 reveals the mechanism behind the generation gap. For unimodal priors, the prior-sampling FID exceeds the reconstruction FID by $\sim$10 points (e.g., 50.09 vs. 39.73 for Gaussian), quantifying how much quality is lost when sampling from $p(\mathbf{z})$ instead of encoding real data. For multimodal priors this gap vanishes: MoG's prior FID (38.46) matches its reconstruction FID (38.76), and VampPrior's prior FID (35.53) is actually *lower* than its reconstruction FID (36.60). This inversion indicates that the learned prior concentrates mass precisely where the aggregate posterior lies, so prior samples traverse the same high-quality

decoder regions as encoded data. This alignment, rather than any improvement in the decoder itself, is the primary driver of the ∼15-point FID reduction.

Sample quality. Figure 1 (Bottom) shows unconditional samples. Gaussian samples are frequently blurry or ambiguous, especially for fine-grained categories (sandals vs. sneakers). VampPrior samples are sharper and more class-consistent, directly aligning with the FID gap. Reconstruction quality is comparable across all priors (Appendix C), confirming differences arise exclusively at sampling time, not in the learned encoder–decoder mapping.

Discussion

Why unimodal priors converge. The Gaussian approximate posterior dominates the aggregate shape: the KL term pulls each $q_\phi(\mathbf{z}|\mathbf{x}_i)$ toward a unimodal, origin-centered target. This overpowers the prior's tail behavior, making tail weight alone an insufficient lever to reorganize latent structure. The active unit counts reinforce this: Gaussian, Laplace, and Student-t all collapse to 8–9 active dimensions despite different tail weights, suggesting the encoder's regularization pressure, not the prior's shape, determines which dimensions survive. A secondary confound is that the non-Gaussian priors use a single-sample Monte Carlo KL estimator (vs. the closed-form Gaussian KL), introducing higher gradient variance; however, the tight convergence of all three unimodal priors across every metric suggests this estimator difference is not a dominant factor.

Why multimodality helps. Multimodal priors alter the regularization topology, letting distinct data clusters occupy separate regions. The prior's support aligns with the aggregate posterior, preventing samples from landing in low-density gaps, explaining the ∼15-point FID drop. VampPrior achieves the best overall results: its pseudo-inputs tightly couple the prior to the encoder, yielding the strongest generation quality across all metrics, while MoG achieves comparable log-likelihood (-231.12 vs. -231.17).

Limitations. Our scope is limited to a single dataset and architecture. Because we fix a Gaussian encoder, results reflect prior–encoder interactions rather than the prior in isolation. We also use a single hyperparameter configuration (latent dim, component count, degrees of freedom) for all priors; ablating these per-prior would strengthen the conclusions but was infeasible within the project timeline.

Conclusion

Our controlled comparison of five VAE priors demonstrates that multimodality, not tail shape, drives generation quality. By improving latent organization without architectural changes, MoG and VampPrior serve as highly effective, drop-in upgrades over the standard Gaussian default. Equally notable, replacing the Gaussian with other unimodal priors (Laplace, Student-t) produces no measurable benefit, suggesting that tail shape alone cannot overcome the regularization pressure of a Gaussian encoder. Our results also highlight a decoupling between log-likelihood and sample quality: MoG and VampPrior achieve near-identical log-likelihoods yet differ by ∼3 FID points, underscoring that density coverage and generation fidelity measure different aspects of model quality.

Acknowledgments

The code for this project is available at https://github.com/HanranSong/VAE.git.

AI tools were used in two capacities. During implementation, AI assistance helped clarify common practices in the VAE literature—including which prior distributions are typically studied and which evaluation metrics are standard and why—as well as the Monte Carlo estimation of KL divergence for non-Gaussian priors, the selection of appropriate evaluation metrics (FID, KID, active units, and importance-weighted log-likelihood), and improvements to the overall code structure. During write-up, AI assistance was used to refine prose clarity, distill key findings from experimental results, and assist with LaTeX layout.

References

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Appendix

A. Reconstruction Metrics

Table 2 reports reconstruction-based FID and KID, computed between real test images and their reconstructions. Unlike the prior-sampling metrics in the main text, these measure encoder–decoder fidelity independent of the prior's sampling distribution. The narrow gap confirms all five models learn similar reconstructions; quality differences arise entirely at sampling time.

Table 2: Reconstruction FID and KID across priors.

Prior	**FID (recon)** $\downarrow$	**KID (recon)** $\downarrow$
Gaussian	39.73	.0220 $\pm$ .0013
Laplace	40.13	.0226 $\pm$ .0012
Student-t	41.06	.0234 $\pm$ .0014
MoG	38.76	.0214 $\pm$ .0014
VampPrior	**36.60**	**.0196** $\pm$ .0011

B. Latent Space Visualizations

Figure 2 shows t-SNE projections for all five priors. The unimodal priors produce similar diffuse embeddings, while MoG and VampPrior show progressively tighter class clusters.

C. Samples and Reconstructions

Figures 3 and 4 show unconditional samples and reconstruction comparisons. Sample quality visibly improves for multimodal priors, while reconstructions remain consistent.

D. Latent Interpolations

Figures 5 and 6 show bilinear interpolations across four Fashion-MNIST classes and within a single class, respectively. All five priors produce plausible transitions, but multimodal priors generate noticeably fewer blurry or ambiguous intermediate images. This is consistent with the latent-space findings: because MoG and VampPrior place prior mass closer to encoded data, linear paths between two latent codes are more likely to pass through well-supported decoder regions rather than low-density gaps.

E. Log-Likelihood vs. Generation Quality

Although MoG and VampPrior achieve nearly identical log-likelihoods (-231.12 vs. -231.17), VampPrior's FID is $\sim$3 points lower (35.53 vs. 38.46). This decoupling suggests that log-likelihood, which measures average density coverage, does not fully predict sample quality. VampPrior's pseudo-inputs are tied to the encoder, so its prior modes track the aggregate posterior as training progresses; MoG's freely parameterized components can spread density more broadly, improving log-likelihood without concentrating mass as tightly on high-quality regions.

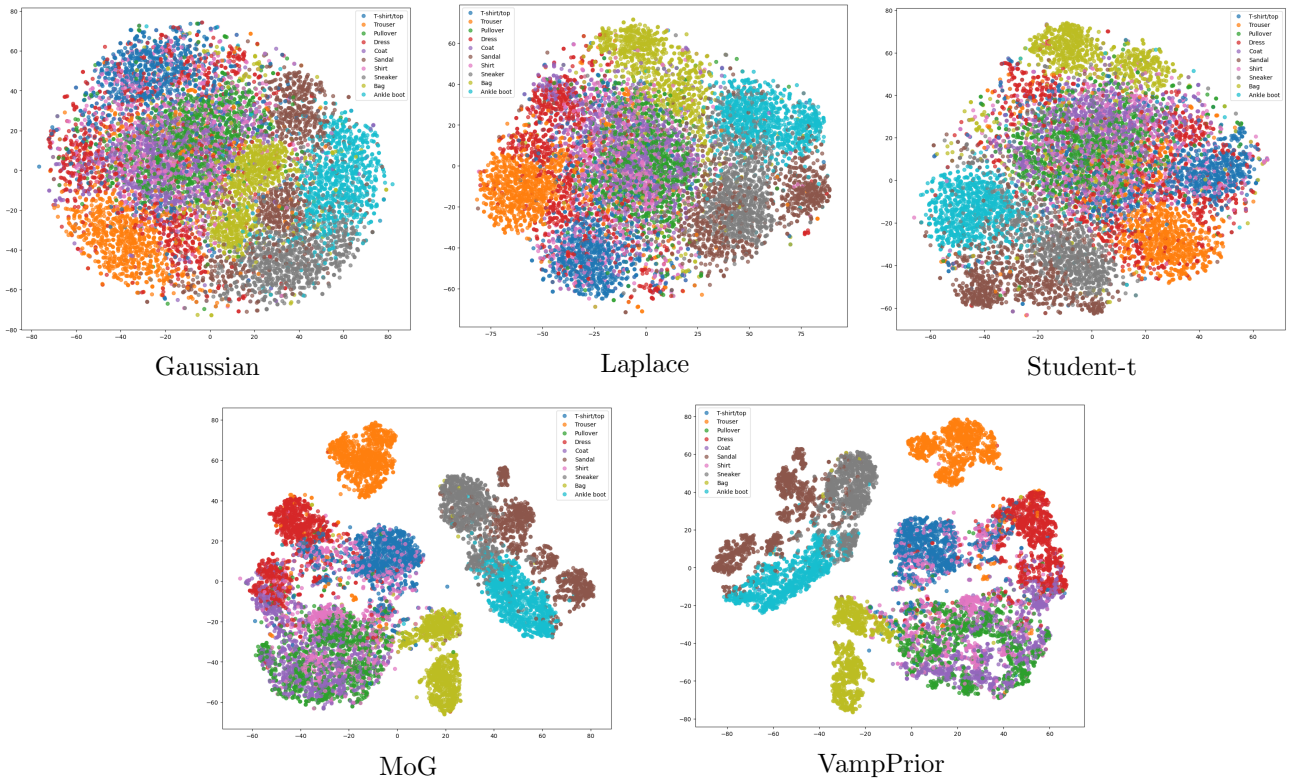

Figure 2: t-SNE projections of test-set latent codes for all five priors. Colors denote Fashion-MNIST classes.

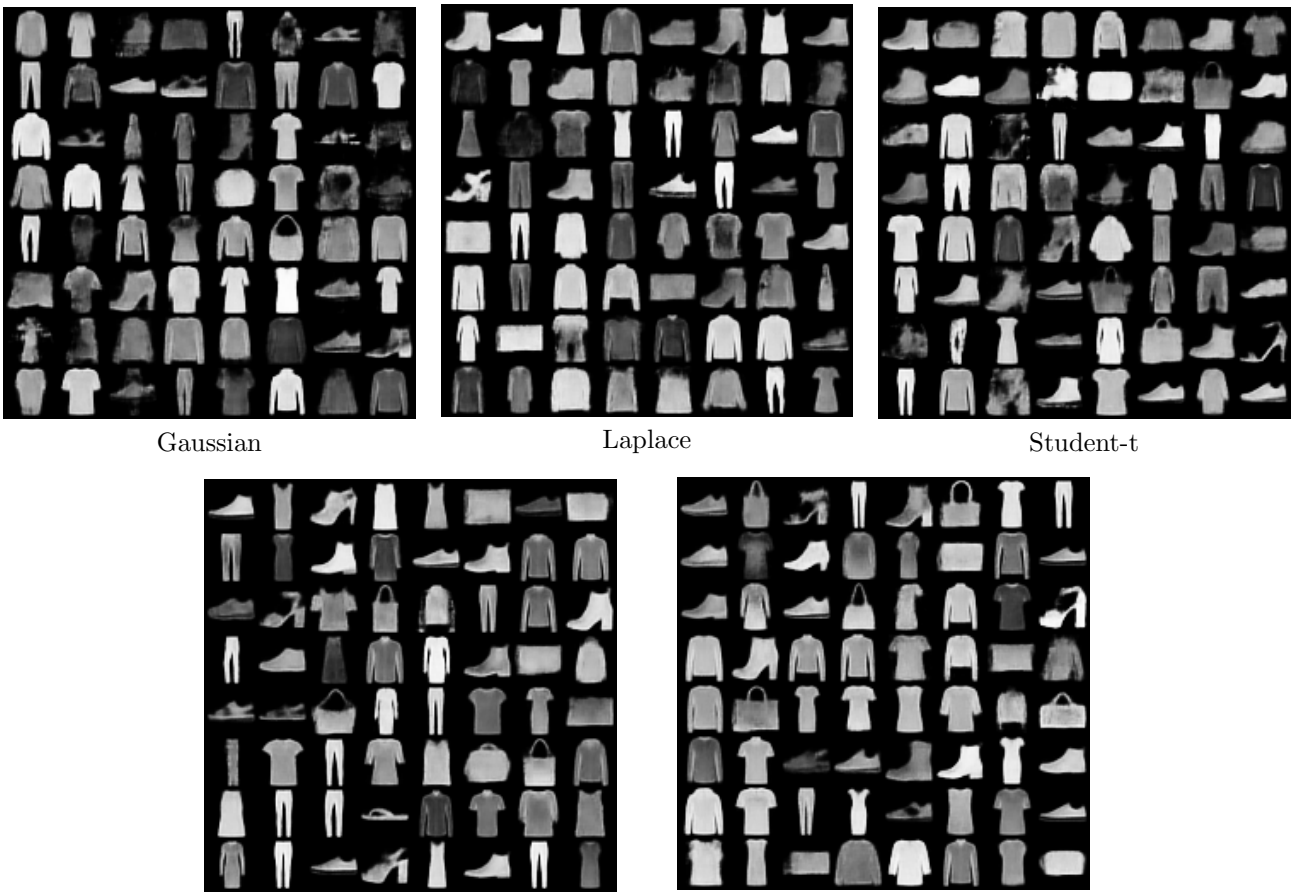

Figure 3: Unconditional samples drawn from the prior.

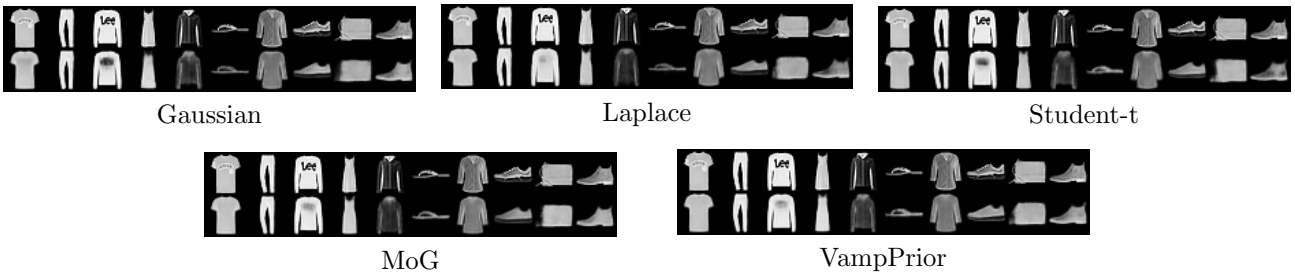

Figure 4: Reconstructions. Within each grid, top rows are original test images; bottom rows are VAE reconstructions.

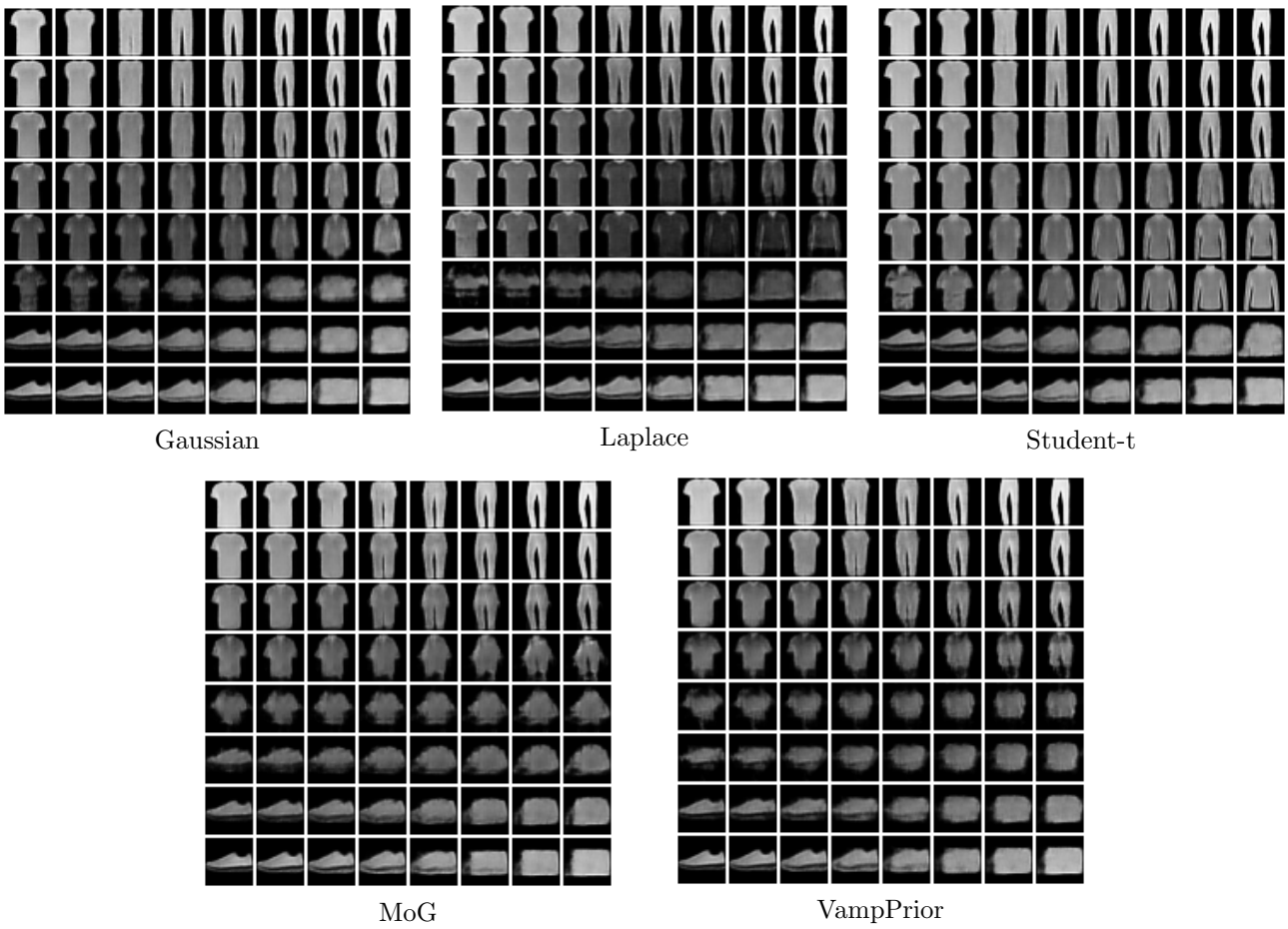

Figure 5: Latent-space bilinear interpolations. Corners are encoded from four distinct Fashion-MNIST classes.

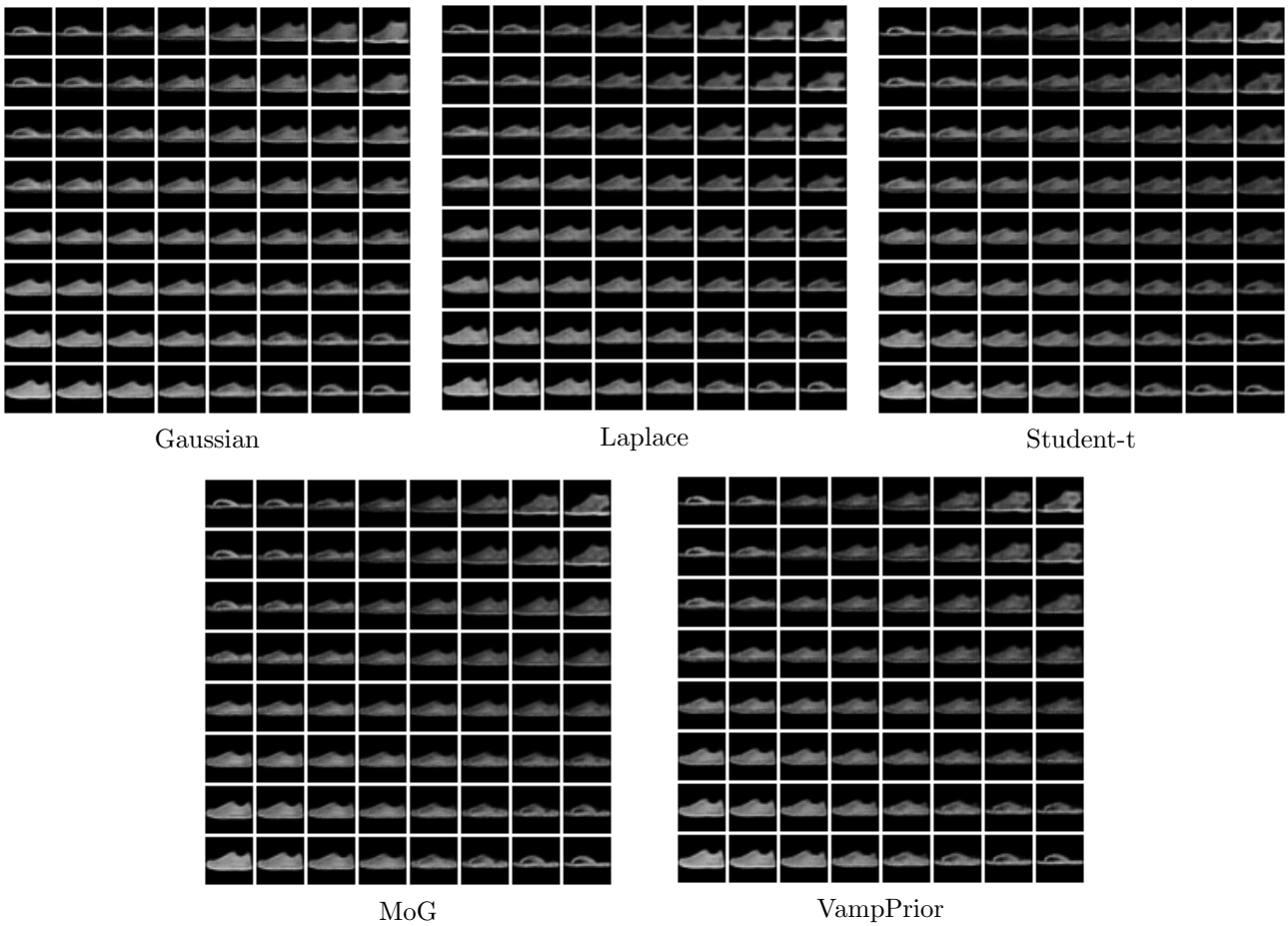

Figure 6: Latent-space bilinear interpolations for a single category. Corners are encoded from four sandal images.